Mathematics Bootcamp

Properties of Preferences and Utility Functions

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Today

- Preference Relations, Indifference, and Utility
- Monotonicity and Local Nonsatiation
- Convexity and Quasiconcavity
- Continuity Through Contour Sets
- Continuous Utility Representation

Preference Relations

- A **preference relation** $\succeq$ is a binary relation on X: a subset of $X \times X$ recording the comparisons made by the DM. It need not compare every pair unless completeness is imposed.
- $x \succeq y$ reads: x is *at least as good* as y (x is *weakly better* than y).
- From $\succeq$ we derive:
  - Strict preference: $x \succ y \iff (x \succeq y$ and not $y \succeq x)$.
  - Indifference: $x \sim y \iff (x \succeq y$ and $y \succeq x)$.

Properties of Indifference

Indifference Relation

If $\succeq$ is rational (i.e., complete and transitive), then the induced indifference relation $\sim$ is:

1. **Reflexive**: $x \sim x$ for all $x \in X$.
2. **Symmetric**: $x \sim y \implies y \sim x$.
3. **Transitive**: $x \sim y \wedge y \sim z \implies x \sim z$.

Equivalence Relation

An equivalence relation is a binary relation that is reflexive, symmetric and transitive.

Equivalence Classes

Equivalence Classes

An equivalence relation induces a **partition** of the set: mutually disjoint subsets whose union equals the entire set.

If $\succeq$ is rational, its induced indifference relation $\sim$

1. is an equivalence relation;
2. generates equivalence classes of indifferent elements.

For each $x \in X$, the equivalence class of x is denoted by

$$[x] = \{y \in X \mid x \sim y\}.$$

Equivalence Classes and Utility

Utility representation

A function $u : X \to \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \quad \Leftrightarrow \quad u(x) \geq u(y).$$

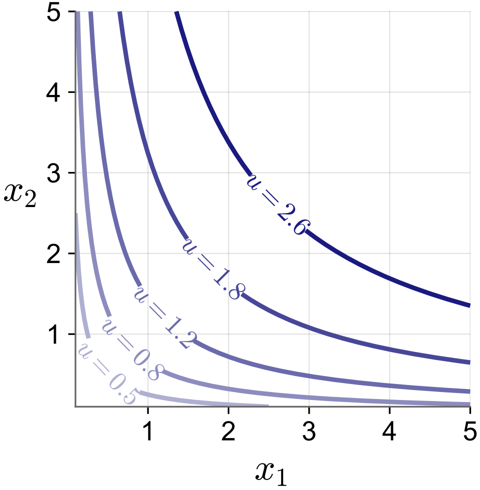
What the Representation Implies

If u represents $\succeq$, then

$$x \sim y \quad \Leftrightarrow \quad u(x) = u(y).$$

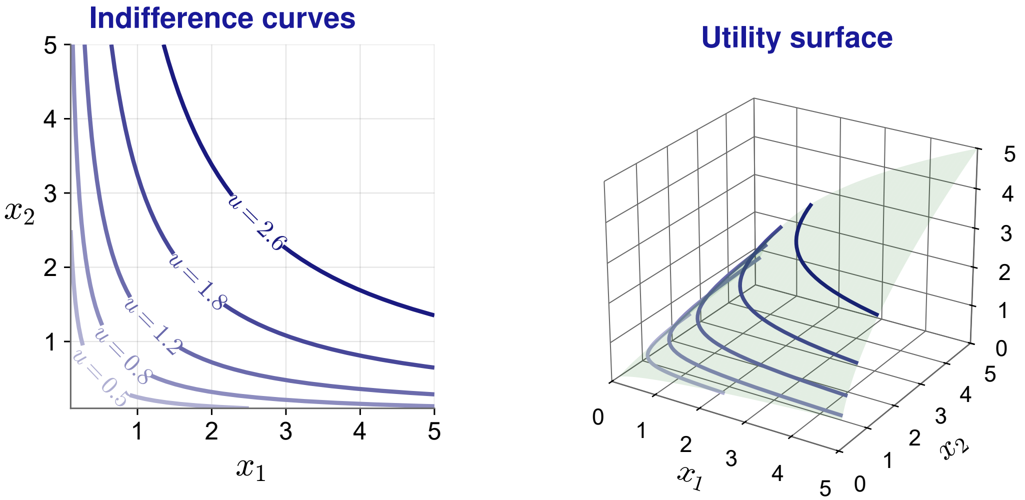
Thus, every equivalence class is mapped to a single utility level. Moreover, if $\phi : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents the same preferences: utility is **ordinal**.

Indifference Curves



- Indifference curves are equivalence classes of X that map to the same utility in $\mathbb{R}$.
- Equivalence classes are disjoint, so indifference curves can never cross.

Utility Representation



A Representation Carries the Structure of Preferences

Property of $\succeq$	**Property of a representation u**
Weak monotonicity	u is non-decreasing.
Strict monotonicity	u is strictly increasing.
Convexity	u is quasiconcave.
Strict convexity	u is strictly quasiconcave.

Continuity is different: a continuous representation gives continuous preferences, while Debreu's theorem gives conditions under which continuous preferences admit at least one continuous representation.

Represent Commodities with Ordered Lists

2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to L (indexed by $\ell = 1, \dots, L$).

As a general matter, a *commodity vector* (or commodity *bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the commodity *space*.[1]

We can use commodity vectors to represent an individual's consumption levels. The ℓth entry of the commodity vector stands for the amount of commodity ℓ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

- Coordinatewise order: $x \geq y$ iff $x_\ell \geq y_\ell$ for every $\ell \in \{1, \dots, L\}$.
- Hence, $x \geq y$ and $x \neq y$ imply $x_\ell > y_\ell$ for some ℓ.

Monotonicity: More is Better

Definitions

Let $X \subseteq \mathbb{R}^L$. A preference relation $\succeq$ on X is

weakly monotone if $x \geq y$ implies $x \succeq y$;

strictly monotone if $x \geq y$ and $x \neq y$ imply $x \succ y$,

for all $x, y \in X$.

A function $f : X \to \mathbb{R}$ is

non-decreasing if $x \geq y$ implies $f(x) \geq f(y)$;

strictly increasing if $x \geq y$ and $x \neq y$ imply $f(x) > f(y)$, for all $x, y \in X$.

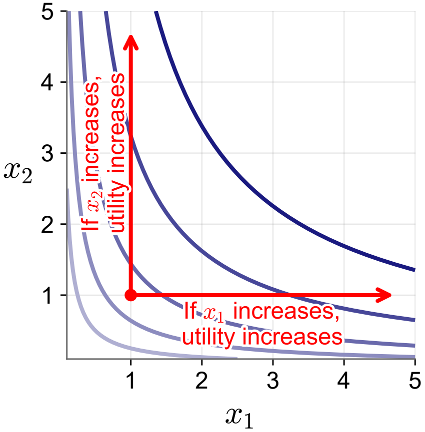
Proposition

If $u : X \to \mathbb{R}$ represents $\succeq$, then

$\succeq$ is weakly monotone *iff* u is non-decreasing;

$\succeq$ is strictly monotone *iff* u is strictly increasing.

Monotonicity: More is Better



Domain: $X = \mathbb{R}^2_{++}$. The function is not strictly increasing on the axes of $\mathbb{R}^2_{+}$.

Local Nonsatiation: No Thick Indifference Curves

Definition: Local Nonsatiation

A preference relation $\succeq$ on X is *locally nonsatiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that $\|x - y\| \leq \varepsilon$ and $y \succ x$.

Thus, preferences are locally nonsatiated if for each x you can find an arbitrarily close y that is strictly better.

Recall that we are in a Euclidean space, so

$$\|x - y\| = \sqrt{\sum_{i=1}^{L} (x_i - y_i)^2}.$$

Local nonsatiation rules out “thick” indifference sets.

Local Nonsatiation: No Thick Indifference Curves

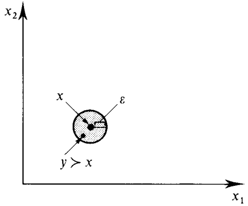


Figure 3.B.1
The test for local nonsatiation.

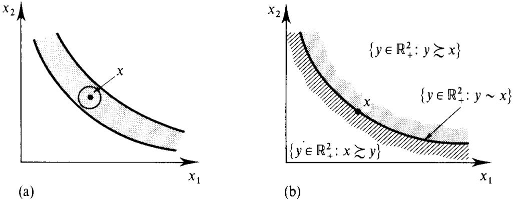


Figure 3.B.2
(a) A thick indifference set violates local nonsatiation.
(b) Preferences compatible with local nonsatiation.

Monotonicity and Local Nonsatiation

Proposition

On $X = \mathbb{R}^L_+$, strict monotonicity implies local nonsatiation.

Proof. Let $x \in \mathbb{R}^L_+$ and $\varepsilon > 0$ be arbitrary.

Define $y = x + \frac{\varepsilon}{L}\mathbf{1}$.

Then $y \in \mathbb{R}^L_+$, $y \geq x$, and $y \neq x$.

Strict monotonicity therefore implies $y \succ x$.

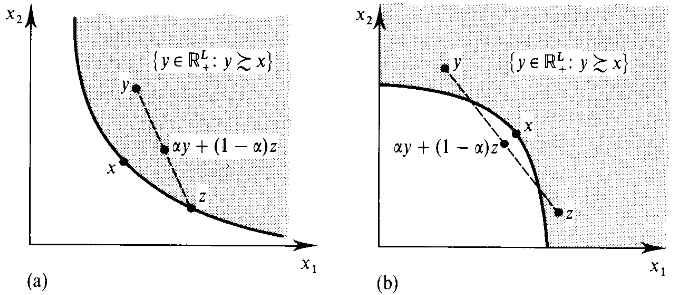
Moreover,

$$\|x - y\| = \sqrt{\sum_{i=1}^{L} \left(\frac{\varepsilon}{L}\right)^2} = \sqrt{L\left(\frac{\varepsilon}{L}\right)^2} = \frac{\varepsilon}{\sqrt{L}} \leq \varepsilon$$

Thus every ε-neighborhood of x contains a strictly better bundle.

However, local nonsatiation does not imply monotonicity.

Convexity: Taste for Variety



Definition 3.B.4: The preference relation $\succsim$ on X is *convex* if for every $x \in X$, the upper contour set $\{y \in X : y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1-\alpha)z \succsim x$ for any $\alpha \in [0, 1]$.

Figure 3.B.3(a) depicts a convex upper contour set; Figure 3.B.3(b) shows an upper contour set that is not convex.

Convexity: Taste for Variety

Convex Sets and Preferences

A set $X \subseteq \mathbb{R}^L$ is *convex* if $x, y \in X$ and $\alpha \in [0,1]$ imply $\alpha x + (1-\alpha)y \in X$.
Let X be convex. For each $x \in X$, define $U(x) := \{y \in X : y \succeq x\}$. A preference relation $\succeq$ on X is *convex* if

$$y, z \in U(x),\ \alpha \in [0,1] \quad \Longrightarrow \quad \alpha y + (1-\alpha)z \in U(x).$$

It is *strictly convex* if $y \neq z$ and $\alpha \in (0,1)$ imply $\alpha y + (1-\alpha)z \succ x$ whenever $y, z \in U(x)$.

Precise Geometric Statement

Convex preferences mean that every upper contour set $U(x)$ is convex. This is the precise statement behind the informal “taste for variety” description.

Level Sets and Quasiconcavity

Definitions

Let $X \subseteq \mathbb{R}^L$ be convex. A function $f : X \to \mathbb{R}$ is *quasiconcave* if, for $x, y \in X$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1-\alpha)y) \geq \min\{f(x), f(y)\}.$$

It is *strictly quasiconcave* if, for $x \neq y$ and $\alpha \in (0, 1)$,

$$f(\alpha x + (1-\alpha)y) > \min\{f(x), f(y)\}.$$

A More Transparent Definition

A function $f : X \to \mathbb{R}$ is *quasiconcave* if every upper level set of f is convex. That is, $\{\, x \in X \mid f(x) \geq c \,\}$ is convex for every $c \in \mathbb{R}$.

Convex Preferences and Quasiconcave Utility

Exact Equivalence

Let X be convex and let $u : X \to \mathbb{R}$ represent $\succeq$. Then

$$\begin{aligned} \succeq \text{ is convex} \quad &\iff \quad u \text{ is quasiconcave,} \\ \succeq \text{ is strictly convex} \quad &\iff \quad u \text{ is strictly quasiconcave.} \end{aligned}$$

Why

$$U(x) = \{y \in X : y \succeq x\} = \{y \in X : u(y) \geq u(x)\}.$$

Thus upper contour sets are upper level sets of u; replacing weak with strict inequalities gives the second equivalence.

Topology on the Choice Set

Relative Topology on $X \subseteq \mathbb{R}^L$

For $x \in X$ and $\varepsilon > 0$, the relative open ball is

$$B_\varepsilon^X(x) = \{y \in X : \|y - x\| < \varepsilon\}.$$

- A set $G \subseteq X$ is *open relative to* X if every $x \in G$ has some $\varepsilon > 0$ with $B_\varepsilon^X(x) \subseteq G$.
- A set $F \subseteq X$ is *closed relative to* X if $X \setminus F$ is open relative to X.

Why "Relative" Matters

At a boundary point of X—for example, where one commodity is zero—the ball is intersected with the feasible choice set X.

Closed Sets Contain Their Limits

Sequential Test for Closed Sets

Let $F \subseteq X \subseteq \mathbb{R}^L$. The set F is closed relative to X if and only if

$$x_n \in F \text{ for every } n, \qquad x_n \to x \in X \quad \Longrightarrow \quad x \in F.$$

Why This Matters

Closedness says that a set keeps the limits of all its convergent sequences. Applied to contour sets, this turns the topological definition of continuous preferences into a test involving limits of rankings.

Lower and Upper Contour Sets

Contour Sets at x

$$U(x) := \{y \in X \mid y \succeq x\} \qquad L(x) := \{y \in X \mid x \succeq y\}.$$

The indifference set through x is always

$$I(x) := \{y \in X : y \sim x\} = U(x) \cap L(x).$$

Weak and Strict Sections

If $\succeq$ is complete, then

$$\{y \in X : y \succ x\} = X \setminus L(x), \qquad \{y \in X : x \succ y\} = X \setminus U(x).$$

Continuity: Equivalent Tests

Continuity via Contour Sets

Let $X \subseteq \mathbb{R}^L$ and let $\succeq$ be complete. We call $\succeq$ *continuous* if, for every $x \in X$, the weak contour sets $U(x)$ and $L(x)$ are closed relative to X.

Equivalent Open-Set Test

Under completeness, continuity is equivalent to requiring $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ to be open relative to X for every $x \in X$.

Interpretation

Small changes in a bundle cannot cause a strict ranking to reverse abruptly.

Continuity: Rankings Survive Limits

Equivalent Sequence Test

Let $X \subseteq \mathbb{R}^L$ and let $\succeq$ be complete and transitive. Continuity is equivalent to

$$(x_n, y_n) \to (x, y), \quad x_n \succeq y_n \text{ for every } n \quad \Longrightarrow \quad x \succeq y.$$

Equivalent Local Test

Equivalently, strict preference is open in $X \times X$: if $x \succ y$, there exist $\varepsilon_x, \varepsilon_y > 0$ such that

$$x' \in B^X_{\varepsilon_x}(x), \quad y' \in B^X_{\varepsilon_y}(y) \quad \Longrightarrow \quad x' \succ y'.$$

A Continuous Representation Gives Closed Contours

Closed Level Sets

If $f : X \to \mathbb{R}$ is continuous, then for every $c \in \mathbb{R}$ the sets

$$f^{-1}\big([c, \infty)\big) \qquad \text{and} \qquad f^{-1}\big((-\infty, c]\big)$$

are closed relative to X.

Apply This to Utility

If u represents $\succeq$, then

$$U(x) = u^{-1}\big([u(x), \infty)\big), \qquad L(x) = u^{-1}\big((-\infty, u(x)]\big).$$

Therefore, a continuous utility representation implies continuous preferences.

Debreu's Representation Theorem

Continuous Representation

Let $X = \mathbb{R}^L_+$. If $\succeq$ is complete, transitive, and continuous on X, then there exists a *continuous* utility function $u : X \to \mathbb{R}$ such that

$$x \succeq y \iff u(x) \geq u(y).$$

- The theorem starts from continuous preferences, defined by closed contour sets.
- The theorem guarantees at least one continuous representation; it does not make every utility representation continuous.
- Rationality is necessary. On $\mathbb{R}^L_+$, continuity supplies enough additional structure for a continuous representation.

These Assumptions Solve Different Problems

Assumptions	Conclusion
Nonempty compact D; continuous u	$\arg\max_{x \in D} u(x)$ is nonempty.
Convex D; strictly quasiconcave u	There is at most one maximizer.
Local nonsatiation; $D = B(p, I)$ with $p \gg 0$	Every optimum satisfies $p \cdot x = I$.

Existence, uniqueness, and a binding constraint are distinct conclusions. Day 4 studies the mathematics that produces each one.